



UNIVERSITY OF SCIENCE - VNUHCM

FACULTY OF INFORMATION TECHNOLOGY

SOFTWARE TESTING

HW6 - PERFORMANCE TESTING

Software Testing Project Report

Authors:

Cao Uyển Nhi (22127310)

ltthuy22@clc.fitus.edu.vn

Supervisors:

Teacher Trần Duy Hoàng

Teacher Hồ Tuấn Thanh

Teacher Trương Phước Lộc

August 11, 2025

Table of Contents

1	Group Information	1
2	Server / PC Configuration	1
3	Performance Testing Techniques Applied	1
3.1	Test case 1: Load Testing	1
3.2	Test case 2: Spike Testing	2
3.3	Test case 3: Stress Testing	2
4	Applying AI in Performance Testing	3
4.1	Prompt 1 — Scenario Design	3
4.2	Prompt 2 — Groovy Randomization Logic	3
4.3	Prompt 3 — Debugging URI Errors	4
4.4	Prompt 4 — Adding Think Time	4
4.5	Prompt 5 — Spike & Stress Test Configuration	4
5	Youtube Video Link	4
6	Self-Assessment	5

1 Group Information

Group ID: 07

Member Name	Student ID	Assigned Features	Status
Cao Uyên Nhi	22127310	Homepage - Category - Product	Done
Lưu Thanh Thuý	22127410	- SignIn	Done
Nguyễn Phước Minh Trí	22127424	- Catalog	Done
		- Categories	Done
Võ Lê Việt Tú	22127435	- MyProfile	Done
		- Order Management	Done
Trần Thị Cát Tường	22127444	- Contact	Done
		- Category Management	Done

2 Server / PC Configuration

Component	Specification
OS	macOS Sonoma 14.x
CPU	Apple M1, 8 cores
RAM	16 GB
Disk	512 GB SSD
JMeter Version	Apache JMeter 5.6.x
Java Version	OpenJDK 17
Browser	Chrome (for script recording)
Website Under Test	The Toolshop (Localhost, Angular front-end + backend API)

3 Performance Testing Techniques Applied

I applied **three types of performance tests**:

- **Load Testing** – Determine if the application can handle the expected number of concurrent users under normal conditions.
- **Spike Testing** – Measure the system's behavior under sudden and extreme load increases.
- **Stress Testing** – Identify the breaking point and observe how the system behaves beyond normal operational capacity.

3.1 Test case 1: Load Testing

Objective: Verify whether the system can handle the expected normal traffic of **50 concurrent users** without performance degradation.

User Journey:

1. Access the home page (/)

2. Select a category (randomized from `category.csv`)
3. Select a product (randomized from `product.csv`)
4. View the product details page

Thread Group Configuration:

Parameter	Value
Number of Threads	50
Ramp-up Period	250 sec
Loop Count	Infinite (or fixed for short test)
Assertions	Response time $\leq$ 2000 ms, HTTP status = 200

Expected Result:

- Stable response times within 2 seconds
- Zero or negligible error rate

3.2 Test case 2: Spike Testing

Objective: Assess the system's reaction and recovery time when traffic suddenly increases.

User Journey: Same as in the Load Testing scenario.

Ultimate Thread Group Configuration:

Stage	Virtual Users	Ramp-up	Hold Load	Shutdown
Warm-up	10	30 sec	60 sec	5 sec
Spike	100	30 sec	60 sec	5 sec
Recovery	10	10 sec	60 sec	5 sec

Expected Result:

- Slight increase in response times during spike, but within 3000 ms
- Near-zero error rate
- System returns to normal performance after load decreases

3.3 Test case 3: Stress Testing

Objective: Identify the maximum load capacity and observe behavior beyond the system's limits.

User Journey: Same as in the Load Testing scenario.

Ultimate Thread Group Configuration:

Stage	Virtual Users	Ramp-up	Hold Load	Shutdown
1	60	10 sec	60 sec	5 sec
2	80	10 sec	60 sec	5 sec

Stage	Virtual Users	Ramp-up	Hold Load	Shutdown
3	120	10 sec	60 sec	5 sec
4	160	10 sec	60 sec	5 sec
5	200	10 sec	120 sec	10 sec

Expected Result:

- Response time and error rate noticeably increase when load exceeds ~180 virtual users
- At 200 virtual users, the system shows instability or failures

4 Applying AI in Performance Testing

During the performance testing process, AI tools were utilized to enhance efficiency, accuracy, and reporting quality. The key applications include:

4.1 Prompt 1 — Scenario Design

Objective: Generate realistic performance testing scenarios for The Toolshop with limited database accounts.

Prompt Example: “When conducting performance testing, what scenarios can be created for The Toolshop given that user registration is not possible and only two accounts exist in the database?”

AI Output:

Suggested focusing on **view-only scenarios**:

- Browse categories
- View products in category
- View product details

Recommended adding load patterns for:

- Load Testing
- Spike Testing
- Stress Testing

4.2 Prompt 2 — Groovy Randomization Logic

Objective: Automate random category and product selection from CSV data for JMeter scripts.

Prompt Example: “Write a JSR223 (Groovy) script in JMeter that reads `categories.csv` and `products.csv`, picks a random category slug, and then picks a random product in that category.”

AI Output:

Provided Groovy code using:

- `FileReader` + `CSVParser` to read files

- Filtering products by category ID
- Storing selected values in JMeter variables (`vars.put("cat_slug", ...)`, `vars.put("product_id", ...)`)

Advised **not to use Random Controller** if randomization is already handled in Groovy.

4.3 Prompt 3 — Debugging URI Errors

Objective: Fix `java.net.URISyntaxException` when using JMeter with dynamic URLs.

Prompt Example: “I get `Illegal character in fragment` for `http://localhost:4200/#/category` in JMeter. How to fix?”

AI Output:

- Explained that `#` fragments are not sent to the server and must be handled differently.
- Suggested switching to actual REST API endpoints instead of SPA routes for backend performance measurement.

4.4 Prompt 4 — Adding Think Time

Objective: Simulate realistic user behavior with pauses between requests.

Prompt Example: “How to add think time between requests in JMeter without breaking my randomization logic?”

AI Output:

- Recommended **Constant Timer** or **Uniform Random Timer** after HTTP Samplers.
- Suggested placing timers inside the Thread Group or directly after each request for scenario realism.

4.5 Prompt 5 — Spike & Stress Test Configuration

Objective: Tune JMeter parameters for different performance testing types.

Prompt Example: “Given my load test uses 50 users, 30s ramp-up, 1 min hold, what should I set for spike testing and stress testing?”

AI Output:

- Spike Test: Sudden jump to peak load (e.g., $0 \rightarrow 100$ users in 5s), short hold.
- Stress Test: Gradual ramp beyond expected limits (e.g., +20 users every 30s until failure).

5 Youtube Video Link

[Load Testing Process](#)

[Spike Testing Process](#)

6 Self-Assessment

Criteria	Description	Max Points	Self Assessment
Load testing	Report: 1.0 TestCases, BugReport: 0.5 Script, Data: 0.5 Video: 1.0	3.0	3.0
Stress testing	Report: 1. TestCases, BugReport: 0.5 Script, Data: 0.5 Video: 1.0	3.0	3.0
Spike testing	Report: 1. TestCases, BugReport: 0.5 Script, Data: 0.5 Video: 1.0	3.0	3.0
Use of AI Tools	Prompt transparency, critical validation, added value	1.0	1.0
Total		10.0	10.0